



AIM-390

AIR PERMEABILITY APPARATUS



Aimil Ltd.

Instrumentation & Technologies

Version

2.0



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**AIRPERMEABILITY
APPARATUS
(Blaine Type)**

Operating Instructions



Operating Instructions

Air Permeability Apparatus, Blaine Type

Introduction

The equipment meets the requirements of IS:5516. It carries the ISI Certification Mark for quality and is a variable flow type air permeability apparatus. It is used to determine the fineness of cements, pozzolanas, and other powdery materials, in terms of the specific surface expressed as the total surface area in square centimeters per gram of material, as per the procedure described in IS:1727, 4031, 4825, ASTM 6204 and BS:4359 (Part II, Appendix 'A').

Description

The equipment is illustrated in the General Assembly Drawing attached. The numbers given against the components of the equipment in the description below pertain to this General Assembly Drawing.

It consists of a permeability cell (4), a perforated disc (5), a manometer (2) and other accessories.

Permeability Cell (4)

The permeability cell consists of a rigid cylinder with the top of the cell at right angles to the principal axis of the cell. The bottom of the cell forms an air-tight connection with the top of the manometer. A ledge forms an integral part of the cell for the support of the perforated metallic disc. The disc is perforated with equally distributed holes and snugly fits inside the cell.

Plunger (6)

The plunger which fits into the cell has sharp square edges. The flat width on the side of the plunger acts as an air vent. The top of the plunger is provided with a collar such that when the plunger is placed in the cell and the collar brought into contact with the top of the cell, the distance between the bottom of the plunger and the top of the perforated disc is 15 ± 1 mm.



Note: A variable flow type air permeability apparatus consists essentially of a means for drawing a definite quantity of air under a falling head through a prepared bed of powder of a definite porosity. The number and size of pores in a prepared bed of cement of definite porosity is a function of the size of particles, and determines the rate of flow through the bed.

Manometer (2)

The top of one arm of the U-Tube manometer is fitted with a rubber stopper for an air tight connection with the permeability cell. The manometer arm connected to the permeability cell has lines etched around the tube and below the top of the side outlet. A side outlet with a stop cock is provided near the top for creating a vacuum in the manometer. The manometer is mounted firmly on a wooden stand in such a manner that the arms are vertical.

Accessories

One rubber cork (3), one tube (7), twelve filter discs, a suction bulb, a punch and dibutylphthalate liquid in a bottle (8) are supplied as accessories.

Setting Up

Place the perforated disc (5) inside the cell (4) resting on the ledge. Insert the plunger (6) into the cell. Fix the cell on the rubber cork (3) on top of the manometer arm and open the stop cock to atmosphere. Fill the manometer with dibutylphthalate to the midpoint and close the stop-cock.

Calibration

Take a standard sample of cement at room temperature. The standard sample of cement can be purchased from National Council for Cement & Building Materials, Ballabgarh on request. Determine the bulk volume of the compacted bed of powder by the mercury displacement method as follows :

Place two filter paper discs in the permeability cell, pressing the edges down with a rod slightly smaller than the diameter of the cell until the filter discs are flat on the perforated metal disc. Apply oil to the interior of the cell, then fill the cell with mercury, removing any air bubbles adhering to the wall of the cell. Level the mercury with the top of the cell by means of a small glass plate. Remove the mercury from the cell, weigh and record the weight of the mercury. Remove one of the filter discs from the cell. Place 2.80 g of cement (it is not necessary to use the standard sample for the bulk volume determination) in the



cell and tap the bed of cement. Place a filter paper disc on top of cement bed and insert the plunger. Compress it until the plunger collar is in contact with the top of the cell and then remove the plunger slowly. (The prepared bed of cement shall be firm. If it is too loose, or if the cement cannot be compressed to the desired volume, adjust the trial quantity of cement used). Fill the remaining space in the top of the cell with mercury, remove entrapped air, and level the top off as before. Remove the mercury from the cell, weigh the mercury and record the value.

Calculate the bulk volume occupied by the cement to the nearest 0.005 cm^3 from the formula:

$$V = \frac{W_a - W_b}{D} \quad \dots (1)$$

Where

V = Bulk volume of cement in cm^3

W_a = Grams of mercury required to fill cell, with no cement in cell.

W_b = Grams of mercury required to fill the portion of the cell not occupied by the prepared bed of cement in the cell.

D = Density of mercury at the test temperature, in g/cm^3 .

Make at least two determinations of bulk volume of cement using separate compactions for each determination. The bulk volume value used for subsequent calculations should be the average of two values agreeing within $\pm 0.005 \text{ cm}^3$. Note the temperature in the vicinity of the cell and record it at the beginning and end of the determination.

Having determined the bulk volume of cement, in cm^3 , proceed with the calibration.

Take the contents of a vial of the standard cement sample in a 125 g. jar and shake vigorously for 2 minutes to fluff the cement and break up lumps or agglomerates. Calculate the weight of sample required such that the porosity of the bed of cement is 0.500 ± 0.005 (For values of Porosity of the cement bed, see the table attached).

$$W = PV (1-e) \quad \dots (2)$$



Where

W = Sample required in grams

P = Specific gravity of test sample (for portland cement a value of 3.15 used).

V = Bulk volume of bed of cement, in cm^3 as determined above.

e = Desired porosity of bed of cement (0.500 ± 0.005)

Weigh a sample to the nearest 0.001 g. Keep the perforated disc on the ledge with a filter disc over it. Place the cement in the cell and compress as already described above.

Moisten the rubber cork before connecting the permeability cell. Attach the cell to the manometer tube making sure that an air tight connection has been made and taking care not to jerk or disturb the prepared bed of cement. Determine the efficiency of the connection by stoppering the top of the cell attached to the manometer, partially evacuating one arm of manometer and then closing the stop cock. Any continuous drop in the pressure indicates a leak in the system. Evacuate the air in the arm of the manometer until the liquid reaches the top mark and close the stop-cock. Start a stop-watch as the bottom of the meniscus of the manometer liquid reaches the second mark (next to the top) and stop it as the bottom of the meniscus reaches the third mark (next to the bottom). Record the time interval in seconds. Record the temperature in degrees centigrade.

Make at least three determinations of time of flow on each of three separately prepared beds of the standard sample. The sample may be re-luffed and reused for preparation of the test bed, provided that it is kept dry and all tests are made within 4 hours of the opening of the sample. Use fresh filter paper discs for each determination. Calibration shall be made by the same operator who performs the fineness test.

Recalibration

Prepare a secondary sample and use it as a fineness standard for checking the instrument between regular calibrations with the standard cement sample.

Recalibration is required at periodic intervals to correct the possible wear on plunger or permeability cell. Also, if any loss in manometer fluid occurs, or if a change is made in the type of quality of the filter paper used for the test, recalibration is required.



Specimen Preparation

Use the same weight of cement as was used during calibration except that when determining the fineness of type III cements, or other types of fine ground cements whose bulk for this weight is so great that the ordinary thumb pressure will not cause the plunger collar to contact the top of the cell. The weight of the sample should be that required to produce a test bed having a porosity of 0.530 ± 0.005 . When this method is used for materials other than portland cement, adjust the weight of sample so that a firm hard bed is produced by the compacting process. Compare the sample in the cell as described under calibration.

Test Procedure

Use the test sample instead of the standard cement samples and proceed as described under calibration

Calculations

The calculation of specific surface value shall be made according to the following formula:

$$S = \frac{S_s \sqrt{T}}{\sqrt{T_s}} \quad \dots (3)$$

$$S = \frac{S_s \sqrt{n_s} \sqrt{T}}{\sqrt{T_s} \sqrt{n}} \quad \dots (4)$$

$$S = \frac{S_s (1 - e_3) \sqrt{e^3} \sqrt{T}}{\sqrt{e^3_s} \sqrt{T_s} \sqrt{(1 - e)}} \quad \dots (5)$$

$$S = \frac{S_s (1 - e_3) \sqrt{e^3} \sqrt{n} \sqrt{T}}{\sqrt{e^3_s} \sqrt{T_s} \sqrt{e^3} \sqrt{(1 - e)}} \quad \dots (6)$$

$$S = \frac{S_s p_s (1 - e_3) \sqrt{e^3} \sqrt{T}}{p (1 - e) \sqrt{e^3_s} \sqrt{T_s}} \quad \dots (7)$$



$$S = \frac{S_s p_s (1 - e_s) \sqrt{n_s} \sqrt{e^3} \sqrt{T}}{p (1 - e) \sqrt{e^3} \sqrt{N}} \quad \dots (8)$$

Where

S = Specific surface in cm^2/g of the test sample

S_s = Specific surface in cm^2/g of the standard sample used in the calibration of the apparatus → ~~4.000000~~

T = Measured time interval, in seconds, of the manometer drop for the test sample

T_s = Measured time interval, in seconds, of the manometer drop for the standard sample used in the calibration of the apparatus. ~~4.0000~~

n_s = Viscosity of air in poises at the temperature of test of the standard sample used in the calibration of the apparatus

n = Viscosity of air in poises at the temperature of test of the test sample.

e_s = Porosity of the prepared bed of standard sample used in the calibration of the apparatus.

e = Porosity of the prepared bed of the test sample.

p_s = Specific gravity of the standard sample used in the calibration of the apparatus.

p = Specific gravity of the test sample (for Portland cement, a value of 3.15 may be used).

Equations 7 and 8 shall be used for the calculation of the fineness of materials other than portland cement. Eq. 7 shall be used when the test temp. of the sample is within $+ 3^\circ \text{C}$ of the temperature of calibration test of the standard fineness sample, and Eq. 8 is used if temperature of tests is outside of this range.

Note: Values for $\sqrt{n}$, $\sqrt{e^3}$ and $\sqrt{T}$ may be taken from Table 1, 2 and 3 respectively.

~~3.15~~
4.00
4.00



Equations 3 and 4 shall be used in calculations of fineness of Portland cements compacted to the same porosity as the standard fineness sample. Equation 3 is used if the temperature of test of the test sample is within $\pm 3^{\circ}\text{C}$ of the temperature of the calibration test of the standard fineness sample, and Equation 4 is used if the temperature of test is outside this range.

Equations 5 and 6 shall be used in the calculation of the fineness of portland cements compacted to some porosity other than that of the standard fineness sample used in the calibration test. Equation 5 is used if the temperature of that test of the test sample is within $\pm 3^{\circ}\text{C}$ of the temperature of calibration test of the standard fineness sample, and equation 6 is used if the temperature of the test is outside this range.

Maintenance

1. All parts of the equipment should always be kept clean.
2. Care should be taken that the holes of the perforated disc do not get blocked.
3. If the manometer liquid used forms a stain, the manometer should be cleaned periodically.

TABLE 1

Density of Mercury, Viscosity of Air (n) and $\sqrt{n}$ at given temperature.

Room Temperature $^{\circ}\text{C}$	Density of Mercury g/cm^3	Viscosity of Air, n , Poises	$\sqrt{n}$
(1)	(2)	(3)	(4)
16	13.56	0.0001788	0.01337
18	13.55	0.0001798	0.01341
20	13.55	0.0001808	0.01344
22	13.54	0.0001818	0.01348
24	13.54	0.0001828	0.01352
26	13.53	0.0001837	0.01355
28	13.53	0.0001847	0.01359
30	13.52	0.0001857	0.01362
32	13.52	0.0001867	0.01366
34	13.51	0.0001876	0.01369

The density of the mercury shall be determined separately.



TABLE 2

Values of Porosity of Cement Bed.

Porosity of Bed, e	$\sqrt{e^3}$
0.495	0.348
0.496	0.349
0.497	0.350
0.498	0.351
0.499	0.352
0.500	0.354
0.501	0.355
0.502	0.356
0.503	0.357
0.504	0.358
0.505	0.359
0.506	0.360
0.507	0.361
0.508	0.362
0.509	0.363
0.510	0.364
0.525	0.380
0.526	0.381
0.527	0.383
0.528	0.384
0.529	0.385
0.530	0.386
0.531	0.387
0.532	0.388
0.533	0.389
0.534	0.390
0.535	0.391

TABLE 3

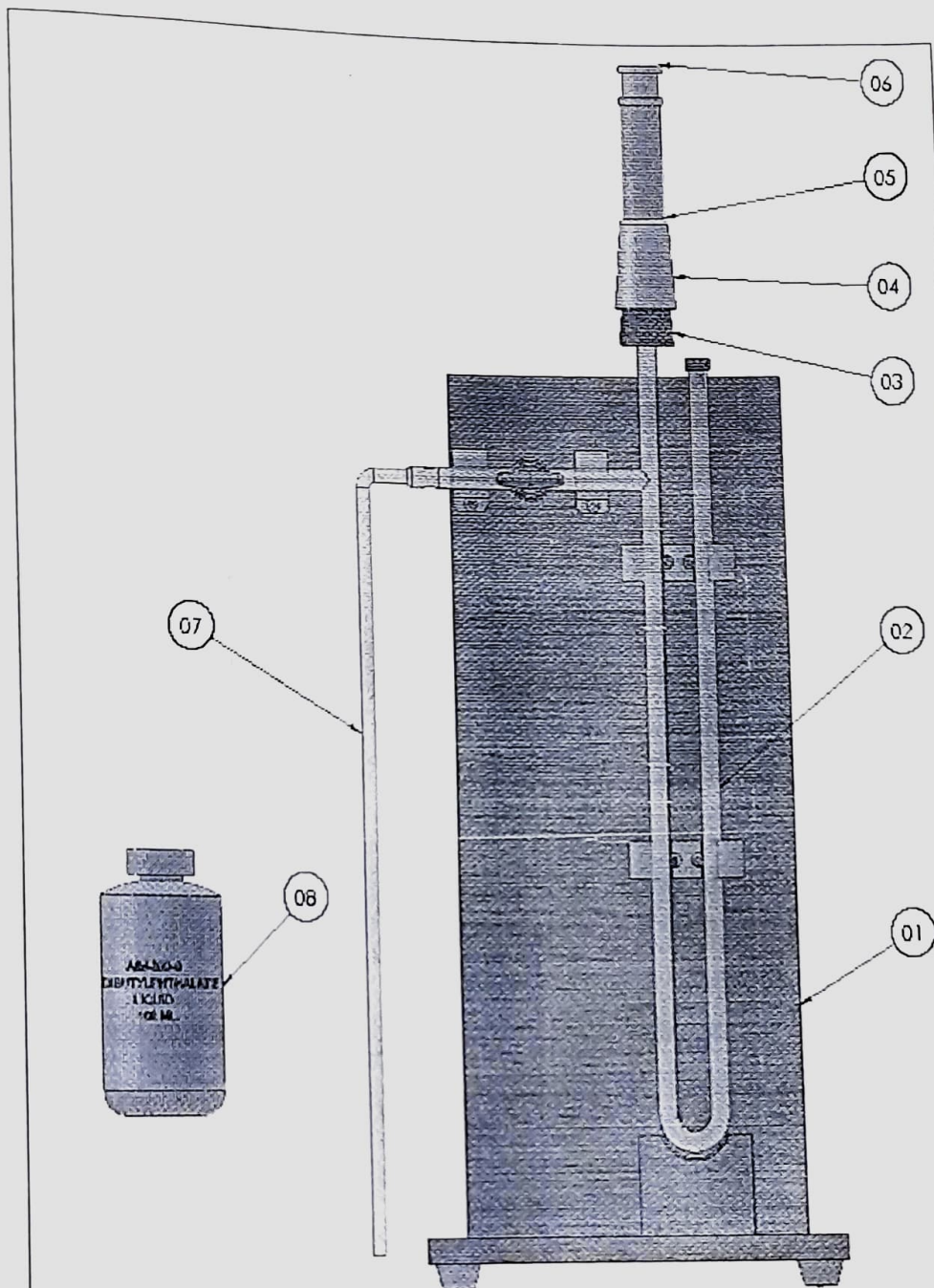
TIME OF AIR FLOW

T = time of air flow in seconds, and
 $\sqrt{T}$ = the factor for use in the equations

T	$\sqrt{T}$	T	$\sqrt{T}$	T	$\sqrt{T}$
26	5.10	28.5	5.34	31	5.57
26.5	5.15	29	5.39	31.5	5.61
27	5.20	29.5	5.44	32	5.66
27.5	5.25	30	5.48	32.5	5.70
28	5.29	30.5	5.52	33	5.74
33.5	5.79	36	6.00	38.5	6.20
34	5.83	36.5	6.04	39	6.24
34.5	5.87	37	6.08	39.5	6.28
35	5.92	37.5	6.12	40	6.32
35.5	5.96	38	6.16	40.5	6.36
41	6.40	43.5	6.60	46	6.78
41.5	6.44	44	6.63	46.5	6.82
42	6.48	44.5	6.67	47	6.86
42.5	6.52	45	6.71	47.5	6.89
43	6.56	45.5	6.74	48	6.93



T	$\sqrt{T}$	T	$\sqrt{T}$	T	$\sqrt{T}$
48.5	6.96	51	7.14	53.5	7.31
49	7.00	51.5	7.18	54	7.35
49.5	7.04	52	7.21	54.5	7.38
50	7.07	52.5	7.24	55	7.42
50.5	7.10	53	7.28	55.5	7.45
56	7.48	58.5	7.65	61	7.81
56.5	7.51	59	7.68	61.5	7.84
57	7.55	59.5	7.71	62	7.87
57.5	7.58	60	7.75	62.5	7.90
58	7.62	60.5	7.78	63	7.94
63.5	7.96	66	8.12	68.5	8.28
64	8.00	66.5	8.16	69	8.31
64.5	8.03	67	8.19	69.5	8.34
65	8.06	67.5	8.22	70	8.37
65.5	8.09	68	8.25	70.5	8.40



AIM-390 AIR PERMEABILITY APPARATUS (BLAINE TYPE)

S.NO.	DESCRIPTION.
01	WOOD STAND
02	MANOMETER
03	RUBBER
04	PERMEABILITY CELL

S.NO.	DESCRIPTION
05	PERFORATED DISC
06	PLUNGER
07	TUBE
08	DIBUTYLPHthalate